



Lab on Geographic Information Science for
sustainable development and smart cities



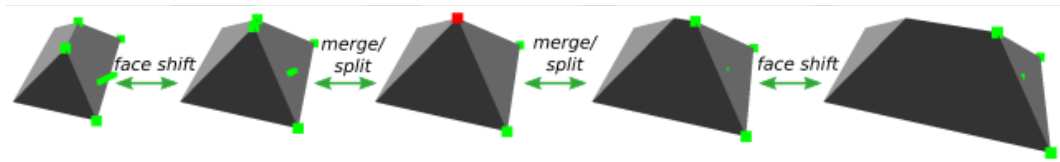
Université
Gustave Eiffel

Polyhedral Modeler Using Face Shifts With Automatic Topologic Updates, Edge Flips and Exact Math based Geometry

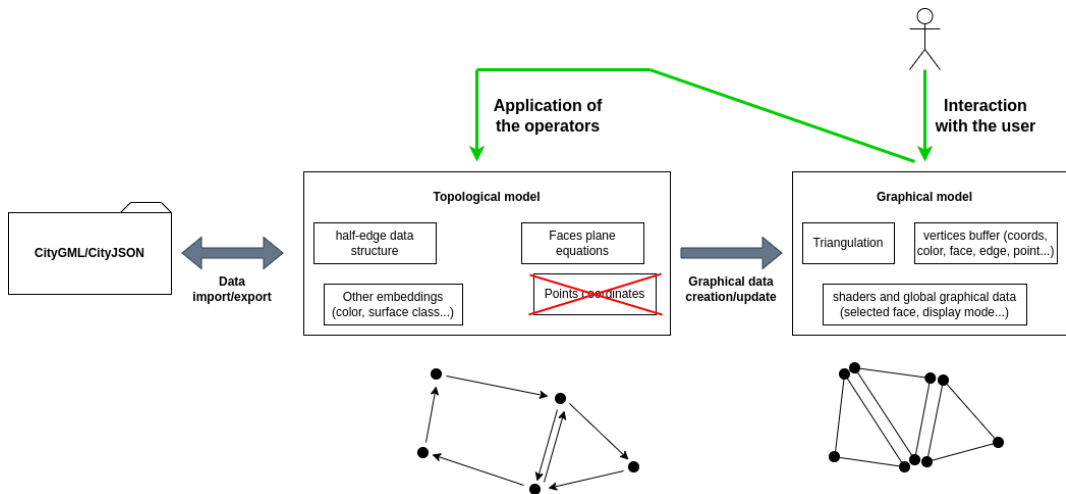
Florent GENIET

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March 11, 2025

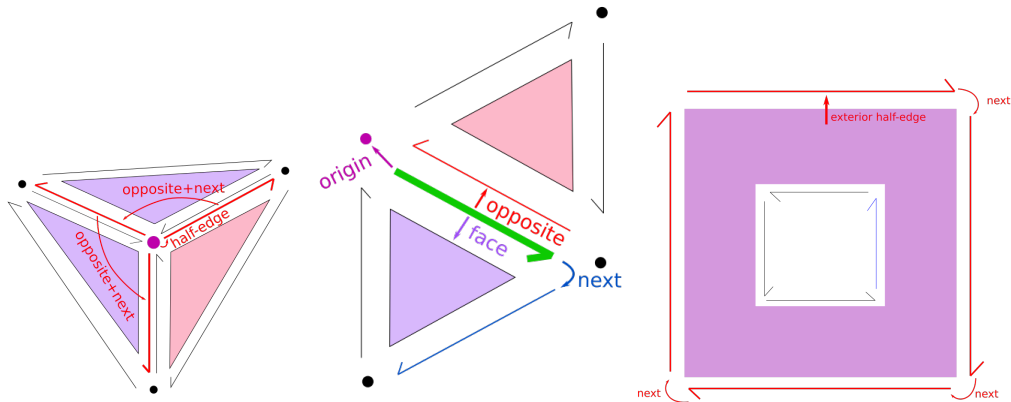
Context : Creation of a polyhedral modeler



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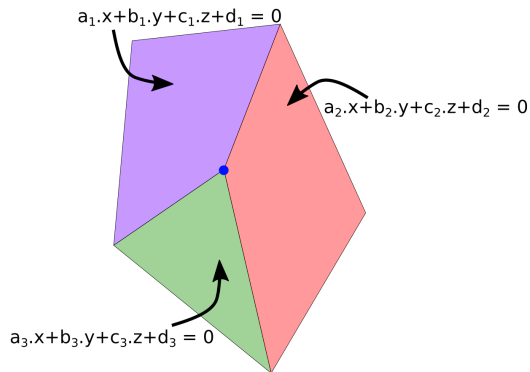


Data structure : Topological data



Data structure : Geometrical data

The **geometrical data** is embedded in the faces, as plane equation coefficients.

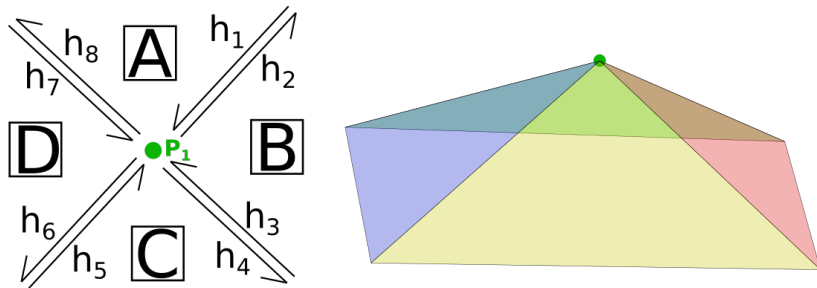


The point coordinates are the solution of the equation system :

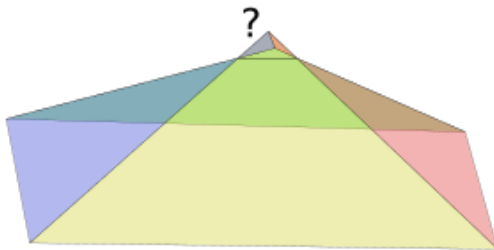
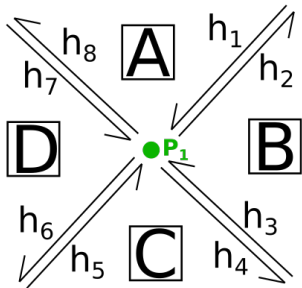
$$\begin{cases} a_1x + b_1y + c_1z + d_1 = 0 \\ a_2x + b_2y + c_2z + d_2 = 0 \\ a_3x + b_3y + c_3z + d_3 = 0 \\ \dots \end{cases}$$

Coherence of the model

Issue : The geometrical and topological data must be coherent.



Coherence of the model



Accuracy issue with floating points numbers

The geometrical coordinates are stored as **floating points** numbers.

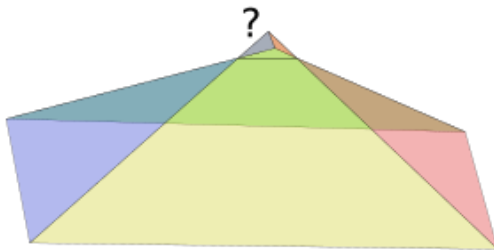
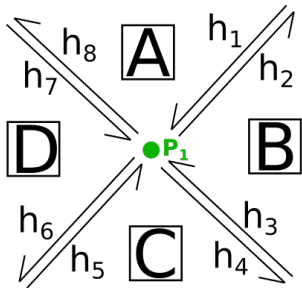
Norm IEEE 754 : $\underbrace{s}_{\text{sign}} \mid \underbrace{e_1 \dots e_8}_{\text{exponent}} \mid \underbrace{f_1 \dots f_{23}}_{\text{fraction}} \rightarrow (-1)^s \cdot 2^e \cdot 1.f$.

Accuracy when changing the fraction : $2^{-23} \cdot 2^{e-127} \geq 2^{-150}$

Accuracy when changing the exponent : $+1.f$ and $-\frac{1.f}{2}$

⇒ There is an imprecision on the coordinates of the plane equations stored in our geometrical data.

Coherence of the model



Exact computation libraries

Basic solution : restrain computations to $\mathbb{Q}$.

⇒ Can not use operators like $\sqrt{}$, $\log$, e^x ...

⇒ Input may have the same issue of imprecision → it needs to be corrected.

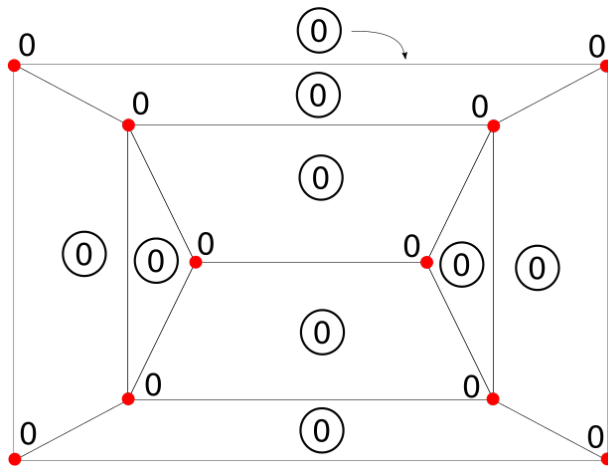
Input data correction algorithm

We use an algorithm to correct the input plane equations according to the topology.

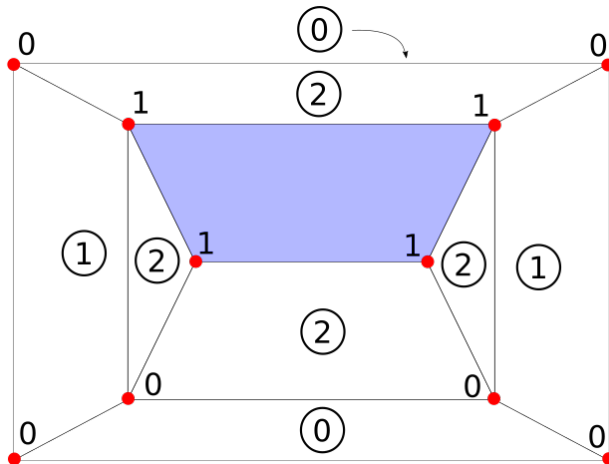
We only change the 4th coordinate of the plane equations :

$$(a, b, c, d) \rightarrow (a, b, c, d \pm \delta)$$

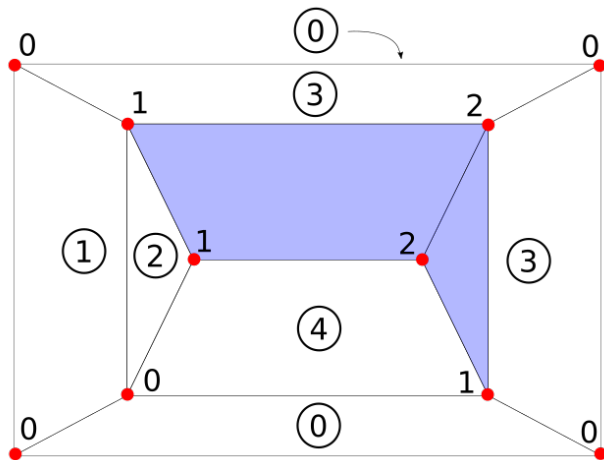
Input data correction algorithm



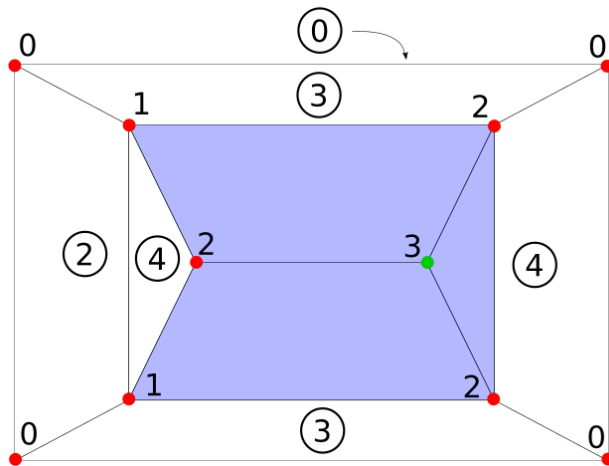
Input data correction algorithm



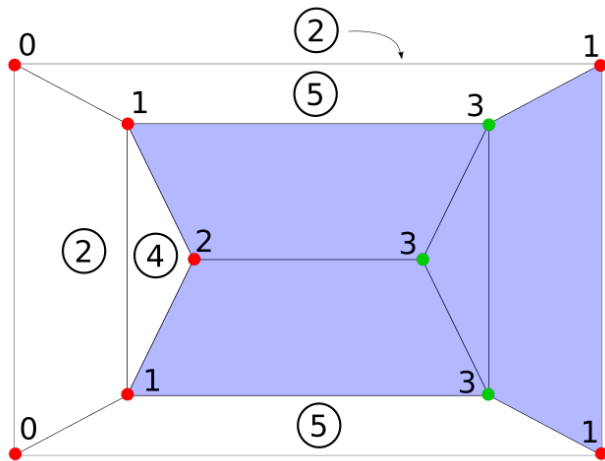
Input data correction algorithm



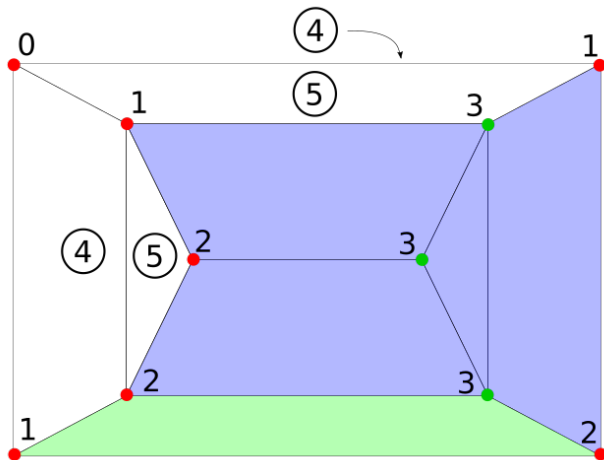
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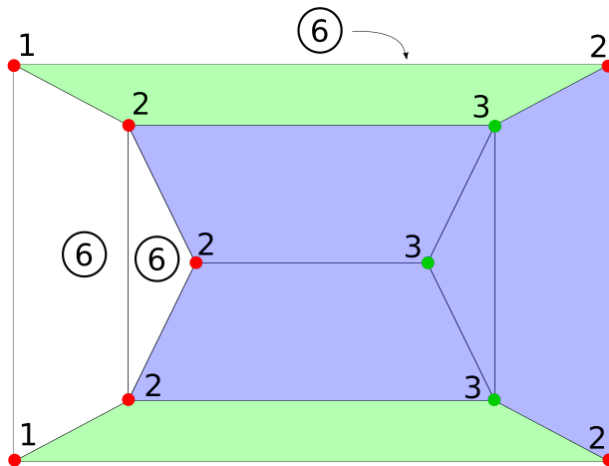
Input data correction algorithm



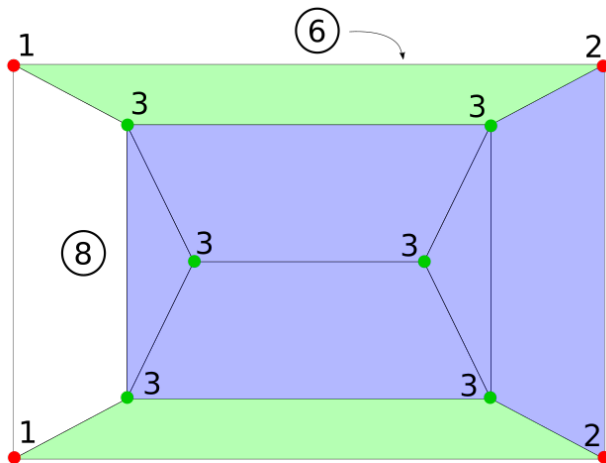
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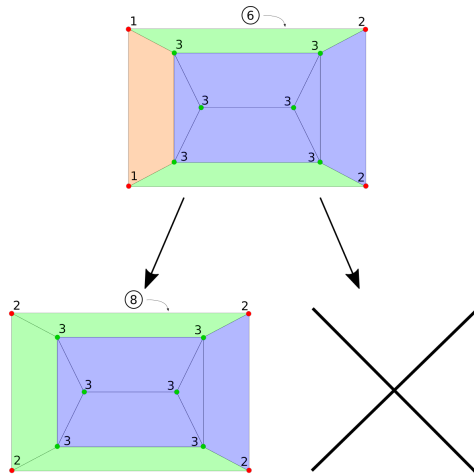
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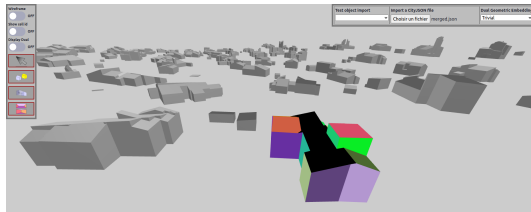


Input data correction algorithm



Conclusion

Import of a cityJSON file, with **95% of successful imports** (460 over 486 buildings successfully imported).



The process can be time consuming for big buildings → only apply it when the user select a building.